

TITLE

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Lemma 0.0.1. Let R be a ring with involution (Definition 1.0.5). Suppose that there is an isomorphism $R \cong \prod_{i=1}^n R_i$ of involution rings and write $e_1, \dots, e_n$ for the complete central (Definition 1.0.1) orthogonal idempotents corresponding to this decomposition. Let $\varepsilon \in Z(R)$ satisfy $\varepsilon \cdot \bar{\varepsilon} = 1$ and write $\varepsilon_i \in R_i$ for the component of ε in R_i . (♠ TODO: orthogonal sum)

1. Every ε -Hermitian form (Definition 1.2.1) $H : V \times V \rightarrow R$ on a (right) R -module V uniquely decomposes as an orthogonal sum $H = \perp_{i=1}^n H_i$ where each $H_i : V_i \times V_i \rightarrow R_i$ is an ε_i -Hermitian form on the R_i -module $V_i = e_i V$. In fact, H_i is constructed as $H|_{V_i} : V_i \times V_i \rightarrow R$ and has values in R_i .
2. If V is a finitely generated projective R -module (equivalently each V_i is a finitely generated projective R_i -module), then H is nonsingular (Definition 1.2.3) if and only if each H_i is nonsingular.

Proof. It suffices to prove this in the case that $R \cong R_1 \times R_2$. We show that claimed bijection has an inverse map that is the map sending a Hermitian form $H : V \times V \rightarrow R$ to the pair $(H|_{V_1}, H|_{V_2})$.

Write $e_i \in R_i$ for complete orthogonal idempotents of $R_1 \times R_2$. Given a ε -Hermitian form $H : V \times V \rightarrow R$, note that $H|_{V_1} : V_1 \times V_1 \rightarrow R$ are respectively valued in R_1 respectively

because

$$\begin{aligned}
H|_{V_1}(v, w) &= H(e_1v, e_1w) \\
&= \overline{e_1} \cdot e_1 H(v, w) \\
&= e_1 \cdot e_1 H(v, w) \\
&= e_1 H|_{V_1}(v, w)
\end{aligned}$$

where $e_1 \in R_1 \times R_2 \cong R$. In fact, $H|_{V_1}$ is a ε_1 -Hermitian form over R_1 because

$$\begin{aligned}
H|_{V_1}(v, w) &= H(e_1v, e_1w) \\
&= e_1 H(e_1v, e_1w) \\
&= e_1 \varepsilon \overline{H(e_1w, e_1v)} \\
&= \varepsilon_1 \overline{H|_{V_1}(w, v)}.
\end{aligned}$$

Similarly, $H|_{V_2} : V_2 \times V_2 \rightarrow R$ is valued in R_2 and is a ε_2 -Hermitian form over R_2 . Moreover, $V \cong V_1 \oplus V_2$ as R -modules, and for any $v_1 \in V_1$ and $v_2 \in V_2$,

$$\begin{aligned}
H(v_1, v_2) &= H(e_1v_1, e_2v_2) \\
&= \overline{e_1} \cdot e_2 H(v_1, v_2) \\
&= e_1 \cdot e_2 H(v_1, v_2) \\
&= 0,
\end{aligned}$$

so $H = H|_{V_1} \perp H|_{V_2}$.

On the other hand, given $H_i : V_i \times V_i \rightarrow R_i$ we show that $H_i : V_i \times V_i \rightarrow R_i \hookrightarrow R_1 \times R_2 \cong R$ is a ε -Hermitian form over R . Given $a \in R$, let $a_i \in R_i$ be the R_i -component of a . For all $v, w \in V_i$,

$$\begin{aligned}
H(v, w) &= \varepsilon_i \overline{H_i(w, v)} \\
&= \varepsilon \overline{H_i(w, v)}.
\end{aligned}$$

For any $v, w \in V_i$ and $a, b \in R$, we then have

$$\begin{aligned}
H_i(av, bw) &= H_i(a_i v, b_i w) \\
&= \overline{a_i} H_i(v, w) b_i \\
&= \overline{a} H_i(v, w) b,
\end{aligned}$$

so H_i is indeed a Hermitian form over R . Note that $(H_1 \perp H_2)_{V_i} = H_i$, so the claimed inverse map is indeed the inverse map to the claimed bijection as desired. By ??, the H_i are all nonsingular if and only if $\perp_i H_i$ is nonsingular. $\square$

1. SESQUILINEAR AND HERMITIAN FORMS

Definition 1.0.1 (Center of a ring). Let R be a (not necessarily commutative) ring. The *center of R* , denoted by $Z(R)$, is the subset of elements in R that commute with every element of R :

$$Z(R) := \{z \in R \mid zr = rz \text{ for all } r \in R\}.$$

The center $Z(R)$ is a commutative subring of R .

Elements of $Z(R)$ are called *central elements of R* .

1.0.1. *Involution ring.* (♠ TODO: don't want involution; want dagger objects)

Definition 1.0.2. Let $\mathcal{C}$ be a category.

A *dagger object in $\mathcal{C}$* is an object X equipped with a functor $(\cdot)^* : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$

together with an isomorphism $i : X \rightarrow X^*$ such that the following diagram commutes:

$$X \xrightarrow{i} X^* \xrightarrow{i^*} X^{**} \cong X = \text{id}_X$$

is an automorphism $i : X \rightarrow X$ such that $i \circ i = \text{id}_X$.

Definition 1.0.3. Let $\mathcal{C}$ be a category. An *involution on an object $X \in \text{Ob}(\mathcal{C})$* is an automorphism $i : X \rightarrow X$ such that $i \circ i = \text{id}_X$.

Definition 1.0.4. Let $\mathcal{C}$ be a category. The *trivial involution on an object C of $\mathcal{C}$* simply refers to the identity map on C ; note that it is an involution.

Definition 1.0.5. Let R be a unital associative ring. An *involution on R* is a anti-ring homomorphism of order 2, i.e. a map $\sigma : R \rightarrow R$ satisfying the following conditions for all $a, b \in R$:

1. $\sigma(a + b) = \sigma(a) + \sigma(b)$;
2. $\sigma(ab) = \sigma(b)\sigma(a)$ (anti-homomorphism property);
3. $\sigma(1) = 1$;
4. $\sigma(\sigma(a)) = a$.

In many contexts, an involution may simply be notated by $\cdot \mapsto \bar{\cdot}$. A *ring with involution* is a ring equipped with an involution.

Equivalently, a ring with involution is a dagger object (Definition 1.0.2) in the category of rings.

Definition 1.0.6. A *homomorphism/morphism $\varphi : (A, \sigma_A) \rightarrow (B, \sigma_B)$ of involution rings* is a ring homomorphism $\varphi : A \rightarrow B$ such that $\sigma_B(\varphi(a)) = \varphi(\sigma_A(a))$ for all $a \in A$.

Definition 1.0.7. Let R be a ring.

The *trivial involution on R* refers to the identity ring homomorphism on R . (♠ TODO: have to change this to a discussion of daggers, not categorical involutions) Equivalently, the trivial involution on R is simply the trivial involution (Definition 1.0.4) on R considered as an object of the category of rings.

Definition 1.0.8. 1. Let (R, σ_R) be a ring with involution (Definition 1.0.5). An *R -algebra with involution* is an R -algebra A equipped with an involution (Definition 1.0.5) σ_A , which may be called an *R -linear involution*, such that:

$$\sigma_A(ra) = \sigma_A(a)\sigma_R(r) \quad \text{for all } r \in R, a \in A.$$

(Equivalently, $\sigma_A(ar) = \sigma_R(r)\sigma_A(a)$).

2. Let R be a ring.

An *R -algebra with R -linear involution* or simply an *R -algebra with involution* usually refers to an R -algebra with involution in the above sense where R is considered as an involution ring equipped with the trivial involution (Definition 1.0.7).

1.1. Sesquilinear forms on modules over involution rings.

Definition 1.1.1. Let (A, σ) be a ring with involution (Definition 1.0.5). Let M be a right A -module. A *σ -sesquilinear form* (or simply *sesquilinear form*) on M is a map $b : M \times M \rightarrow A$ such that for all $u, v, w \in M$ and $a \in A$:

1. $b(u + v, w) = b(u, w) + b(v, w)$ and $b(u, v + w) = b(u, v) + b(u, w)$;
2. $b(ua, v) = \sigma(a)b(u, v)$ (conjugate-linear in the first variable);
3. $b(u, va) = b(u, v)a$ (linear in the second variable).

Alternatively, one can impose the convention that the module M be a left module and that the sesquilinear form is so that the first variable is linear and the second variable is conjugate-linear. The above convention is used in [?].

We may denote the set of sesquilinear forms on M by $\text{Sesq}_A(M)$. It is a module over the center of A (Definition 1.0.1).

Definition 1.1.2. Let (A, σ) be a ring with involution (Definition 1.0.5). The construction $M \mapsto \text{Sesq}_A(M)$ (Definition 1.1.1) is functorial; there is a contravariant functor

$$\text{Sesq}_A : \text{Mod}_A^{\text{op}} \rightarrow {}_{Z(A)}\text{Mod}$$

(Definition 1.0.1) that sends a homomorphism $\varphi : M_1 \rightarrow M_2$ of A -modules to the $Z(A)$ -linear map

$$\text{Sesq}_A(\varphi) : \text{Sesq}_A(M_2) \rightarrow \text{Sesq}_A(M_1), \quad \text{Sesq}_A(\varphi)(b)(x, y) = b(\varphi(x), \varphi(y)).$$

Equivalently, the corresponding map

$$\text{Hom}_A(M_2, M_2^*) \rightarrow \text{Hom}_A(M_1, M_1^*)$$

for the adjoints is given by $h \mapsto \varphi^* h_\varphi$ (Definition 1.1.14).

Definition 1.1.3. Let (A, σ) be a ring with involution (Definition 1.0.5). Let (M_1, b_1) and (M_2, b_2) be (right) modules equipped with sesquilinear forms (Definition 1.1.1). A *morphism* $(M_1, b_1) \rightarrow (M_2, b_2)$ between the sesquilinear modules is a (right) A -module homomorphism such that $\text{Sesq}(\varphi)(b_2) = b_1$, i.e.

$$b_2(\varphi(x), \varphi(y)) = b_1(x, y)$$

for all $x, y \in M_1$.

Definition 1.1.4. Let (A, σ) be a ring with involution (Definition 1.0.5). Let (M_1, b_1) and (M_2, b_2) be (right) modules equipped with sesquilinear forms (Definition 1.1.1).

An *isometry* $\varphi : (M_1, b_1) \rightarrow (M_2, b_2)$ is a morphism of sesquilinear modules (Definition 1.1.3) that is an isomorphism. If an isometry exists between the two sesquilinear modules, then the two forms are said to be *isometric*.

Definition 1.1.5. Let (A, σ) be a ring with involution (Definition 1.0.5).

The *category of modules with sesquilinear forms over (A, σ)* is the category whose objects are pairs (M, b) where M is a (right) A -module and $b : M \times M \rightarrow A$ is a sesquilinear form (Definition 1.1.1) and whose morphisms are isometries (Definition 1.1.4).

In particular, such a category is a groupoid.

Definition 1.1.6. Let (A, σ) be a ring with involution (Definition 1.0.5). Let (M, b) be (right) module equipped with a sesquilinear form (Definition 1.1.1). The isometries (Definition 1.1.4) of (M, b) form a group with respect to composition called the *orthogonal* or *unitary group of (M, b)* . The group may be denoted by notations such as $O(M, b)$, $O_A(M, b)$, $U(M, b)$, and $U_A(M, b)$.

1.1.1. Twisted dual modules on involution rings.

Definition 1.1.7. Let R be a (not necessarily commutative) ring. Depending on the module structure of M , we define its dual module as follows:

1. If M is a left R -module, then the *(right) dual module of M* is

$$M^* = M^\vee := \text{Hom}_R(M, R) = \text{Hom}_{R-\mathbb{Z}}(M, R).$$

Note that it is a right R -module, as M is a $R-\mathbb{Z}$ -bimodule and R is an $R-R$ -bimodule.

2. If M is a right R -module, then the *(left) dual module of M* is

$${}^*M = {}^\vee M := \text{Hom}_R(M, R) = \text{Hom}_{\mathbb{Z}-R}(M, R).$$

Note that it is a left R -module, as M is a $\mathbb{Z}-R$ -bimodule and R is an $R-R$ -bimodule.

3. If M is a two-sided R -module, then the *dual of M* usually refers to either the right or the left dual as above.

If a convention for the rightness or the leftness of modules is fixed, then the appropriate dual module is often denoted by $M^\vee$.

In any case, the functor $M \mapsto M^\vee$ is a contravariant functor from the appropriate category of modules to itself. More explicitly, the dual module constructions yield functors

$$\begin{aligned} (-)^\vee : {}_R \text{Mod}^{\text{op}} &\rightarrow \text{Mod}_R \\ (-)^\vee : \text{Mod}_R^{\text{op}} &\rightarrow {}_R \text{Mod}. \end{aligned}$$

If R is a field F and V is an F -vector space, then the dual module

$$V^* = V^\vee := \text{Hom}_F(V, F)$$

is called the *dual vector space of V* .

Definition 1.1.8. Let R be a (not necessarily commutative) ring. Let M be a (left/right) R -module.

The *canonical (evaluation) map from M to its double dual* refers to the morphism

$$M \rightarrow (M^\vee)^\vee = \text{Hom}_R(M^\vee, R) = \text{Hom}_R(\text{Hom}_R(M, R), R), \quad m \mapsto (f \mapsto f(m)).$$

Definition 1.1.9. Let R be a ring with involution (Definition 1.0.5) $r \mapsto \bar{r}$.

Let M be a left/right R -module. The *opposite module*, also called the *conjugate module* or the module by restriction of scalars along the involution (not to be confused with the notion of opposite modules for modules over general rings), is the right/left R -module, denoted by notations such as M^{op} or $\overline{M}$, with the same underlying abelian group as M , where the right R -action is defined as follows

•

$$m \cdot r = \bar{r}m$$

for all $m \in M$ and $r \in R$ if M is a left R -module.

$$r \cdot m = m\bar{r}$$

for all $m \in M$ and $r \in R$ if M is a right R -module.

The opposite/conjugate module construction yields functors

$$\begin{aligned} \overline{(-)} : {}_R \text{Mod} &\rightarrow \text{Mod}_R \\ \overline{(-)} : \text{Mod}_R &\rightarrow {}_R \text{Mod}. \end{aligned}$$

Proposition 1.1.10. Let R be a ring with involution (Definition 1.0.5). The opposite module constructions provide an equivalence of categories between the category of left R -modules and the category of right R -modules. In fact, if M is a left/right R -module, then $(M^{op})^{op}$ is naturally isomorphic to M as a left/right R -module via the identity map.

Definition 1.1.11. Let (A, σ) be a ring with involution (Definition 1.0.5). Let N be a right A -module. The *twisted dual module*, denoted by notations such as N^* , (or sometimes $N^\vee$; not to be confused with the standard dual (Definition 1.1.7) $N^\vee$) or $\text{Hom}_A(N, A)_\sigma$, is the set of all A -linear maps $f : N \rightarrow A$, equipped with the left A -module structure given by

$$(f \cdot r)(n) = \sigma(r) \cdot f(n)$$

for all $r \in A$, $f \in N^\vee$, $n \in N$. Equivalently, it may be constructed as $\overline{M^\vee}$, where by $\overline{}$, we mean the opposite module (Definition 1.1.9) and by $\cdot^\vee$, we mean the standard dual (Definition 1.1.7).

Symmetrically, one may also construct the twisted dual module N of a left A -module N ; twisted dual module constructions thus yield functors

$$\begin{aligned} \text{Mod}_A^{op} &\rightarrow \text{Mod}_A \\ {}_A \text{Mod}^{op} &\rightarrow {}_A \text{Mod}. \end{aligned}$$

Given a morphism $h : M \rightarrow N$ of right or left A -modules, we may write h^* for the morphism $h^* : N^* \rightarrow M^*$ in functorial fashion.

The construction $M^* = \overline{M^\vee}$ is naturally isomorphic to $(\overline{M})^\vee$ (Lemma 1.1.12).

Lemma 1.1.12 ([?, Chapter I, 2.1.1 Lemma]). Let A be a ring with involution (Definition 1.0.5) denoted by $a \mapsto \bar{a}$.

For (left/right) A -modules M , the maps

$$T_M : \overline{M^\vee} \rightarrow \overline{M}^\vee$$

(Definition 1.1.7) (Definition 1.1.9) given by

$$f \mapsto \left(T_M(f) : \overline{m} \mapsto \overline{f(m)} \right)_s$$

where $\overline{m} = m$ as an element over $\overline{M}$, defines a natural isomorphism

$$\overline{(-)}^\vee \rightarrow (-)^\vee$$

between the functors $\overline{(-)}^\vee, (-)^\vee$ from the category of (left/right) A -modules to itself.

In other words, the twisted dual module (Definition 1.1.11) M^* may be constructed either as $\overline{M^\vee}$ or as $\overline{M}^\vee$.

Definition 1.1.13 ([?, Chapter I, After Lemma 2.1.1]). Let A be a ring with involution (Definition 1.0.5) denoted by $a \mapsto \bar{a}$.

Given a left/right A -module M , one can speak of the double (twisted) dual module (Lemma 1.1.12) $(M^*)^*$.

By Lemma 1.1.12, there is a canonical isomorphism

$$\overline{T_{M^\vee}} : \overline{(M^\vee)^\vee} \rightarrow M^{**}$$

(Definition 1.1.7) (Definition 1.1.9). Define the *canonical morphism from M to its double (twisted) dual* as the composition

$$\text{can}_M : M \rightarrow (M^\vee)^\vee \cong \overline{(M^\vee)^\vee} \xrightarrow{\overline{T_{M^\vee}}} M^{**}$$

where $M \rightarrow (M^\vee)^\vee$ is the canonical evaluation map (Definition 1.1.8).

More explicitly, $\text{can}_M(m) = m^{**} \in M^{**}$, where $m^{**}(f) = \overline{f(m)}$. Moreover, can_M specifies a natural transformation from the identity functor to the double dual functor on the appropriate category of A -modules:

$$\begin{aligned} \text{can}_M : \text{id}_{{}_A \text{Mod}} &\Rightarrow (-)^{**} : {}_A \text{Mod} \rightarrow {}_A \text{Mod} \\ \text{can}_M : \text{id}_{\text{Mod}_A} &\Rightarrow (-)^{**} : \text{Mod}_A \rightarrow \text{Mod}_A \end{aligned}$$

1.1.2. *Sesquilinear forms $M \times M \rightarrow A$ are equivalent to maps $M \rightarrow M^*$.*

Definition 1.1.14 ([?, Chapter I, 2.2]). Let (A, σ) be a ring with involution (Definition 1.0.5) and M a right/left A -module equipped with a sesquilinear form (Definition 1.1.1) $b : M \times M \rightarrow A$.

We define two A -linear mappings associated with b :

1. The *left adjoint map* is the morphism of A -modules:

$$\hat{b}_l : M \rightarrow M^*, \quad u \mapsto (v \mapsto b(u, v))$$

where $M^* = \text{Hom}_A(M, A)_\sigma$ is the twisted dual module (Definition 1.1.11).

2. The *right adjoint map* is the morphism of A -modules:

$$\hat{b}_r : M \rightarrow M^*, \quad v \mapsto (u \mapsto \sigma(b(u, v)))$$

If b is an ε -hermitian form (Definition 1.2.1), these maps are related by $\hat{b}_l = \hat{b}_r \cdot \varepsilon$ (under the standard identification of the module and its dual).

When we simply talk about the *adjoint map* of b on the right/left A -module M , we will mean the left/right adjoint respectively and denote it by $\hat{b}$ or by h_b .

Definition 1.1.15. Let (A, σ) be a ring with involution (Definition 1.0.5), and let M be a (left) A -module. Let $h : M \rightarrow M^*$ be a (left) A -module homomorphism where M^* is the twisted dual module of M (Definition 1.1.11). The *sesquilinear form associated to h* is the sesquilinear form (Definition 1.1.1) $b_h : M \times M \rightarrow A$ given by $b_h(x, y) = h(x)(y)$.

Lemma 1.1.16 ([?, Chapter I, 2.2]). Let (A, σ) be a ring with involution (Definition 1.0.5), and let M be a (left) A -module. A sesquilinear form (Definition 1.1.1) $b : M \times M \rightarrow A$ on M is determined by its adjoint (Definition 1.1.14) $\hat{b} = h_b$. Conversely, A -linear homomorphisms $b : M \rightarrow M^*$ (Definition 1.1.11) are determined by their associated sesquilinear forms (Definition 1.1.15) b_h . In fact, the map $b \mapsto h_b$ is a (two-sided) $Z(A)$ (Definition 1.0.1)-module isomorphism

$$\text{Sesq}_A(M) \cong \text{Hom}_A(M, M^*).$$

Definition 1.1.17. Let A be a ring with involution (Definition 1.0.5) denoted $\bar{}$ and M a right A -module equipped with a sesquilinear form (Definition 1.1.1) $b : M \times M \rightarrow A$.

Define a sesquilinear form $b^* : M \times M \rightarrow A$, which may be called by names such as the *conjugate transpose of b* or the *adjoint sesquilinear form of b* (not to be confused with the adjoint map $M \rightarrow M^*$ of Definition 1.1.14), by

$$b^*(x, y) = \overline{b(y, x)}, \quad x, y \in M.$$

Lemma 1.1.18 ([?, Chapter I, 2.3]). Let (A, σ) be a ring with involution (Definition 1.0.5) where σ is also denoted by $\bar{}$, let $\varepsilon \in Z(A)$ be an element such that $\varepsilon \cdot \bar{\varepsilon} = 1$, let M be a right A -module, and let $b : M \times M \rightarrow A$ be a sesquilinear form (Definition 1.1.1) on M .

We have

$$h_{b^*} = (h_b)^* \circ \text{can}_M$$

where b^* (Definition 1.1.17) is the conjugate transpose of b , h_b (Definition 1.1.14) is the adjoint of b , can_M (Definition 1.1.13) is the canonical map from M to its double twisted dual, and h_b^* is the map

$$h_b^* : M^{**} \rightarrow M^*$$

induced from $h_b : M \rightarrow M^*$ by the contravariance of the twisted dual (Definition 1.1.11) functor $(-)^*$.

1.2. ε -Hermitian forms on modules over involution rings.

Definition 1.2.1. Let (A, σ) be a ring with involution (Definition 1.0.5) where σ is also denoted by $\bar{\cdot}$, let $\varepsilon \in Z(A)$ be an element such that $\varepsilon \cdot \bar{\varepsilon} = 1$, and let M be a right A -module. A ε -hermitian form on M (or simply *hermitian form on M* , especially when ε is understood or is just 1) is a σ -sesquilinear form (Definition 1.1.1) $\phi : M \times M \rightarrow A$ satisfying the symmetry condition:

$$\phi(y, x) = \varepsilon \cdot \overline{\phi(x, y)}$$

for all $x, y \in M$. The pair (M, ϕ) is often called a *hermitian module*.

1-hermitian forms may simply be called *hermitian forms*, and (-1) -hermitian forms are called *skew-hermitian forms*.

Definition 1.2.2. Let (A, σ) be a ring with involution (Definition 1.0.5), and let $\varepsilon \in Z(A)$ be an element such that $\varepsilon \cdot \bar{\varepsilon} = 1$.

The *category of modules with ε -Hermitian forms over (A, σ)* is the category whose objects are pairs (M, h) where M is a (right) A -module and $h : M \times M \rightarrow A$ is a ε -Hermitian form (Definition 1.2.1) and whose morphisms are isometries (Definition 1.1.4).

In particular, such a category is a groupoid and is a full subcategory of the category of modules with sesquilinear forms over (A, σ) (Definition 1.1.5).

Definition 1.2.3. Let (A, σ) be a ring with involution (Definition 1.0.5) and M a right A -module equipped with a sesquilinear form (Definition 1.1.1) $\phi : M \times M \rightarrow A$. Let

$$\hat{\phi} : M \rightarrow M^*, \quad x \mapsto \phi(x, \cdot)$$

be the induced (left) adjoint map (Definition 1.1.14).

1. The form ϕ is said to be *nondegenerate* if $\hat{\phi}$ is injective.

In the case where ϕ is an ε -hermitian form (Definition 1.2.1), this condition is equivalent to requiring that $\phi(x, y) = 0$ for all $y \in M$ implies $x = 0$.

2. The form ϕ is said to be *nonsingular* or *regular* if $\hat{\phi}$ is an isomorphism.

Definition 1.2.4 ([?, Chapter I, 2.3]). Let A be a ring with involution (Definition 1.0.5) denoted by $\bar{\cdot}$, let $\varepsilon \in Z(A)$ satisfy $\varepsilon \cdot \bar{\varepsilon} = 1$, and let M be a (right) A -module.

The *form symmetrizer map* is the map

$$S_\varepsilon : \text{Sesq}_A(M) \rightarrow \text{Sesq}_A(M), \quad b \mapsto b + \varepsilon b^*,$$

(Definition 1.1.1) (Definition 1.1.17). It corresponds corresponds under the $Z(A)$ (Definition 1.0.1)-module isomorphism

$$\begin{aligned} \text{Sesq}_A(M) &\cong \text{Hom}_A(M, M^*) \\ b &\mapsto h_b \\ b_h &\leftarrow h \end{aligned}$$

(Lemma 1.1.16) (Definition 1.1.14) (Definition 1.1.15) to the *operator symmetrizer map*

$$S_\varepsilon : \text{Hom}_A(M, M^*) \rightarrow \text{Hom}_A(M, M^*), \quad h \mapsto h + \varepsilon \cdot (h^* \circ \text{can}_M),$$

(Definition 1.1.13) where $h^* : M^{**} \rightarrow M^*$ is the morphism induced from $h : M \rightarrow M^*$ by the contravariance of the twisted dual (Definition 1.1.11) functor $(-)^*$. In other words, we have

$$h_{S_\varepsilon(b)} = S_\varepsilon(h_b)$$

(Definition 1.1.14).

Notation 1.2.5 ([?, Chapter I, 2.3]). Let A be a ring with involution (Definition 1.0.5) denoted by $\bar{}$, let $\varepsilon \in Z(A)$ satisfy $\varepsilon \cdot \bar{\varepsilon} = 1$, and let M be a (right) A -module.

Write

$$\begin{aligned} \text{Sesq}_\varepsilon(M) &:= \text{Im}(S_\varepsilon) \subset \text{Sesq}_A(M) \\ \text{Sesq}^\varepsilon(M) &:= \text{Ker}(S_{-\varepsilon}) \subset \text{Sesq}_A(M). \end{aligned}$$

(Definition 1.2.4); in general, $\text{Sesq}_\varepsilon(M) \subset \text{Sesq}^\varepsilon(M)$.

The groups $\text{Sesq}^\varepsilon(M)$ and $\text{Sesq}_\varepsilon(M)$ are the groups of ε -Hermitian forms on M (Definition 1.2.1) and of even ε -Hermitian forms on M respectively.

Definition 1.2.6 ([?, Chapter I, 2.3]). Let A be a ring with involution (Definition 1.0.5), let $\varepsilon \in Z(A)$ satisfy $\varepsilon \cdot \bar{\varepsilon} = 1$, and let M be an (right) A -module. Write $\text{Sesq}_\varepsilon(M)$ and $\text{Sesq}^\varepsilon(M)$ as in Notation 1.2.5.

One calls elements of $\text{Sesq}_\varepsilon(M)$ *even ε -Hermitian forms*; in other words, a sesquilinear form (Definition 1.1.1) $b \in \text{Sesq}_A(M)$ is an even ε -Hermitian form if and only if there exists some sesquilinear form $k \in \text{Sesq}_A(M)$ such that $b = k + \varepsilon k^*$ (Definition 1.1.14).

In general, $\text{Sesq}_\varepsilon(M) \subset \text{Sesq}^\varepsilon(M)$. If 2 is a unit in A , then $\text{Sesq}_\varepsilon(M) = \text{Sesq}^\varepsilon(M)$, since $b = \frac{1}{2}S_\varepsilon(b)$ for $b \in \text{Sesq}^\varepsilon(M)$.

1.3. Sesquilinear and Hermitian modules and spaces.

1.3.1. Sesquilinear and Hermitian modules.

Definition 1.3.1. Let A be a ring with involution (Definition 1.0.5).

A *sesquilinear module over A* is a pair (M, b) with M a finitely generated projective (right) A -module and b a sesquilinear form (Definition 1.1.1) on M .

We may identify b with its adjoint (Definition 1.1.14) and write (M, h) to denote an ε -Hermitian module with $h \in \text{Hom}_A(M, M^*)$ (Definition 1.1.11).

Remark 1.3.2. This terminology is not strictly present in [?].

Definition 1.3.3 ([?, Chapter I, 3.1]). Let A be a ring with involution (Definition 1.0.5), and let $\varepsilon \in Z(A)$ satisfy $\varepsilon \cdot \bar{\varepsilon} = 1$.

1. A *ε -Hermitian module over A* is a sesquilinear module (Definition 1.3.1) (M, b) over A such that the sesquilinear form (Definition 1.1.1) b is in fact a ε -Hermitian form (Definition 1.2.1) on M .

In other words, a ε -Hermitian module over A is a pair (M, b) with M a finitely generated projective (right) A -module and b a ε -Hermitian form (Definition 1.2.1) on M .

As with sesquilinear modules, we may identify b with its adjoint (Definition 1.1.14) and write (M, h) to denote an ε -Hermitian module with $h \in \text{Hom}_A(M, M^*)$ (Definition 1.1.11).

2. An *even ε -hermitian module over A* is a ε -hermitian module whose underlying hermitian form is even (Definition 1.2.6).
3. A *symmetric bilinear form module over A* is an 1-Hermitian module over A where A has the trivial involution.
4. A *skew-symmetric bilinear form module over A* is a (-1) -Hermitian module over A where A has the trivial involution.

Definition 1.3.4 (cf. [?, Chapter I, 3.2]). Let (A, σ) be a ring with involution (Definition 1.0.5), and let $\varepsilon \in Z(A)$ be an element such that $\varepsilon \cdot \bar{\varepsilon} = 1$.

The *category of ε -Hermitian modules over (A, σ)* is the full subcategory $\mathfrak{Herm}^\varepsilon(A)$ of the category of modules with ε -hermitian forms over (A, σ) (Definition 1.2.2) whose objects are the ε -Hermitian modules (Definition 1.3.3), i.e. pairs (M, h) where M is a finitely generated projective (one-sided) A -module.

The *category of even ε -Hermitian modules over (A, σ)* is the full subcategory $\mathfrak{Herm}_+^\varepsilon(A)$ of $\mathfrak{Herm}^\varepsilon(A)$ whose objects are the even (Definition 1.2.6) ε -hermitian modules over A .

In particular, such categories are groupoids.

1.3.2. Sesquilinear and Hermitian spaces.

Definition 1.3.5 (cf. [?, Chapter I, 3.2]). Let (A, σ) be a ring with involution (Definition 1.0.5).

1. A *sesquilinear space* is a sesquilinear module (Definition 1.3.1) (M, b) whose underlying sesquilinear form b is nonsingular (Definition 1.2.3).
2. A *ε -hermitian space* is a ε -hermitian module (Definition 1.3.3) whose underlying Hermitian form is nonsingular (Definition 1.2.3).
3. An *even ε -hermitian space* is a ε -hermitian space (Definition 1.3.3) whose underlying Hermitian form is even (Definition 1.2.6).

In each case, note that a “space” is simply a “module” whose underlying sesquilinear form is nonsingular.

Definition 1.3.6 (cf. [?, Chapter I, 3.2]). Let (A, σ) be a ring with involution (Definition 1.0.5), and let $\varepsilon \in Z(A)$ be an element such that $\varepsilon \cdot \bar{\varepsilon} = 1$.

The *category of ε -Hermitian spaces over (A, σ)* is the full subcategory $\mathfrak{H}^\varepsilon(A)$ of the category of ε -hermitian spaces over (A, σ) (Definition 1.2.2) whose objects are the ε -Hermitian spaces (Definition 1.3.3), i.e. pairs (M, h) where M is a finitely generated projective (one-sided) A -module and h is nonsingular (Definition 1.2.3).

The *category of even ε -Hermitian spaces over (A, σ)* is the full subcategory $\mathfrak{H}_+^\varepsilon(A)$ of $\mathfrak{H}^\varepsilon(A)$ whose objects are the even (Definition 1.2.6) ε -hermitian spaces over A .

In particular, such categories are groupoids.

Proposition 1.3.7. Let (R, σ) be a ring with involution (Definition 1.0.5) and $\phi : M \times M \rightarrow R$ a sesquilinear form (Definition 1.1.1). If M is a finitely generated projective R -module, then the adjoint map $\hat{\phi} : M \rightarrow \text{Hom}_R(M, R)_\sigma$ is an isomorphism if and only if ϕ is non-degenerate (Definition 1.2.3).

1.4. Unitary rings and unitary modules over unitary rings.

Definition 1.4.1 (see [?, Chapter I, 5.1]). Let A be an involution ring (Definition 1.0.5). Let $\varepsilon \in Z(A)$ (Definition 1.0.1) such that $\varepsilon \cdot \bar{\varepsilon} = 1$. Let Λ be an additive subgroup of A such that

(♠ TODO: $S_{-\varepsilon}(A)$, $S^{-\varepsilon}(A)$)

1. $S_{-\varepsilon}(A) \subseteq \Lambda \subseteq S^{-\varepsilon}(A)$
2. $a \cdot x \cdot \bar{a} \in \Lambda$ for all $x \in \Lambda$ and $a \in A$.

The triple $(A, \varepsilon, \Lambda)$ is called a *unitary ring* and the pair (ε, Λ) is called a form parameter for A .

A *morphism of unitary rings* $f : (A, \varepsilon, \Lambda) \rightarrow (A', \varepsilon', \Lambda')$ is a ring homomorphism $f : A \rightarrow A'$ such that $f(\varepsilon) = \varepsilon'$ and $f(\Lambda) \subset \Lambda'$.

Definition 1.4.2 (see [?, Chapter I, 5.1]). Let R be an involution ring (Definition 1.0.5). A *unitary R -algebra* is a unitary ring (Definition 1.4.1) $(A, \varepsilon, \Lambda)$ such that A is an R -algebra with an R -linear involution (Definition 1.0.8) and Λ is an R -submodule of A such that $S_{-\varepsilon}(A) \subseteq \Lambda \subseteq S^{-\varepsilon}(A)$ and $a \cdot x \cdot \bar{a} \in \Lambda$ for all $x \in \Lambda$ and $a \in A$.

(♠ TODO: $S_{-\varepsilon}(A)$, $S^{-\varepsilon}(A)$)

Lemma 1.4.3 (see [?, Chapter I, 5.1]). Let $(R, \varepsilon, \Lambda)$ be a commutative unitary ring (Definition 1.4.1).

For any commutative R -algebra R' , the triple $(A \otimes_R R', \varepsilon' = \varepsilon \otimes 1, \Lambda' = \Lambda \otimes_R R')$ is a unitary R' -algebra (Definition 1.4.2).

Definition 1.4.4. (♠ TODO: Sesq Lambda epsilon)

1.5. Base change of Sesquilinear forms, Hermitian forms, and Unitary modules.

Lemma 1.5.1 ([?, Chapter I, 7.1]). Let $\eta : A \rightarrow A'$ be a homomorphism of rings with involution (Definition 1.0.6). Given a right A -module M and there is a right A' -module homomorphism

$$j_M : M^* \otimes_A A' \rightarrow (M \otimes_A A')^* = \text{Hom}_{A'}(M', A')$$

(Definition 1.1.11) (??) (??) given by

$$j_M(f \otimes a')(m \otimes b') = \bar{a}' \eta(f(m)) b'$$

for $f \in M^*$, and $a', b' \in A'$. The maps j_M specify a natural transformation in the category of right A' -modules.

If M is a projective right A -module, then j_M is an isomorphism.

Definition 1.5.2 ([?, Chapter I, 7.1]). Let $\eta \rightarrow A \rightarrow A'$ be a homomorphism of rings with involution (Definition 1.0.6). Let M be a right A -module and let $b : M \times M \rightarrow A$ be a sesquilinear form (Definition 1.1.1) on M .

By the *base change of b* (which may also be called by names such as *extension of scalars of b* or the *induction of b along η* as per ??) as the sesquilinear form $b \otimes_A A' : (M \otimes_A A') \times (M \otimes_A A') \rightarrow A'$ (denoted by b' or $b \otimes 1_{A'}$ in [?]) on the right A' -module $M \otimes_A A'$ given by

$$(b \otimes_A A')(m \otimes a', n \otimes a'') = \bar{a}' \eta(b(m, n)) a'' \quad \text{for all } m, n \in M, a', a'' \in A'.$$

and extended A' -sesquilinearly.

The pair $(M \otimes_A A', b \otimes_A A')$ may also be written as $(M, b) \otimes_A A'$. (♠ TODO: discuss adjoint of base change)

Lemma 1.5.3. Let R and S be rings (not necessarily commutative), and let $f : R \rightarrow S$ be a ring homomorphism. Let M be a projective right (resp. left) R -module. The base change $M \otimes_R S$ (resp. $S \otimes_R M$) is a projective right (resp. left) S -module.

Proof. A module over a ring is projective if and only if it is a direct summand of a free module. In particular M is a direct summand of a free R -module, so $M \otimes_R S$ is a direct summand of a free S -module and hence is a projective S -module. $\square$

Lemma 1.5.4. Let $\eta : A \rightarrow A'$ be a homomorphism of rings with involution (Definition 1.0.6). Let M be a right A -module and let $b : M \times M \rightarrow A$ be a sesquilinear form (Definition 1.1.1) on M .

1. If (M, b) is a sesquilinear module (Definition 1.3.1) over A , then $(M, b) \otimes_A A'$ is a sesquilinear module over A' .
2. Let $\varepsilon \in Z(A)$ satisfy $\varepsilon \cdot \bar{\varepsilon} = 1$. If b is a ε -hermitian form (Definition 1.2.1) over A , then $b \otimes_A A'$ is an $\eta(\varepsilon)$ -hermitian form (Definition 1.2.1) over A' . Moreover, if b is even (Definition 1.2.6), then so is $b \otimes_A A'$.

Proof. 1. Since a sesquilinear module is simply a projective module equipped with a sesquilinear form, this follows from Lemma 1.5.3
 2. To show that $b \otimes_A A'$ is an $\eta(\varepsilon)$ -hermitian form, it suffices to verify

$$(b \otimes_A A')(m \otimes a', n \otimes a'') = \eta(\varepsilon) \overline{(b \otimes_A A')(n \otimes a'', m \otimes a')}$$

for all $m, n \in M$ and $a', a'' \in A'$. The left equals $\overline{a'}\eta(b(m, n))a''$, whereas the right equals

$$\begin{aligned}
\eta(\varepsilon)\overline{\overline{a''}\eta(b(n, m))}a' &= \eta(\varepsilon)\overline{a'}\eta(\overline{b(n, m)})\overline{a''} \\
&= \eta(\varepsilon)\overline{a'}\eta(\overline{\overline{b(n, m)}})a'' \\
&= \eta(\varepsilon)\overline{a'}\eta(\overline{\varepsilon\overline{b(m, n)}})a'' \\
&= \eta(\varepsilon)\overline{a'}\eta(\overline{\overline{b(m, n)}} \cdot \overline{\varepsilon})a'' \\
&= \eta(\varepsilon)\overline{a'}\eta(b(m, n)\overline{\varepsilon})a'' \\
&= \overline{a'}\eta(\varepsilon\overline{\varepsilon}b(m, n))a'' \\
&= \overline{a'}\eta(b(m, n))a''
\end{aligned}$$

(♠ TODO: prove evenness)

□

1.6. Transfer of sesquilinear forms via a morphism of involution rings.

Definition 1.6.1 (see [?, Chapter I, 7.2]). Let $\eta : A \rightarrow A'$ be a homomorphism of rings with involution (Definition 1.0.6). Given a right A' -module M' , write $M'_{(A)}$ for the restriction of M' as a right A -module (via η). Let $\tau : A' \rightarrow A$ be a morphism of A -bimodules.

τ induces an additive map

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)})$$

(Definition 1.1.1) that sends a sesquilinear form $b' \in \text{Sesq}_{A'}(M')$ on an A' -module M' to the sesquilinear form $\tau(b') \in \text{Sesq}_A(M'_{(A)})$ on $M'_{(A)}$ constructed by

$$\tau(b')(x, y) = \tau(b'(x, y)).$$

It may be appropriate to call the induced map τ the *transfer* or *Scharlau transfer on sesquilinear forms*

Lemma 1.6.2 (see [?, Chapter I, 7.2]). Let $\eta : A \rightarrow A'$ be a homomorphism of rings with involution (Definition 1.0.6). Given a right A' -module M' , write $M'_{(A)}$ for the restriction of M' as a right A -module (via η). Let $\tau : A' \rightarrow A$ be a morphism of A -bimodules and assume that

$$\tau(\overline{a}) = \overline{\tau(a)}$$

for all $a \in A'$

The Scharlau transfer (Definition 1.6.1)

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)})$$

(Definition 1.1.1) restricts to maps

$$\tau : \text{Sesq}^{\varepsilon'}(M') \rightarrow \text{Sesq}^{\varepsilon}(M'_{(A)})$$

$$\tau : \text{Sesq}_{-\varepsilon'}(M') \rightarrow \text{Sesq}_{-\varepsilon}(M'_{(A)})$$

(Notation 1.2.5)

Definition 1.6.3 (see [?, Chapter I, 7.2]). Let $\eta : (A, \Lambda, \varepsilon) \rightarrow (A', \Lambda', \varepsilon')$ be a homomorphism of unitary rings (Definition 1.4.1). Write $\bar{\cdot}$ for both of the involutions on A and A' . Let $\tau : A' \rightarrow A$ be a morphism of A -bimodules and assume that

$$\tau(\bar{a}) = \overline{\tau(a)}$$

for all $a \in A'$.

The Scharlau transfer (Definition 1.6.1)

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)})$$

(Definition 1.1.1) restricts to a map

$$\tau : \text{Sesq}_{\Lambda'}^{\varepsilon'}(M') \rightarrow \text{Sesq}_{\Lambda}^{\varepsilon}(M'_{(A)})$$

(♠ TODO: Sesq) It is appropriate to also call this induced τ the *Scharlau transfer for sesquilinear forms on modules over unitary rings*

Definition 1.6.4 (see [?, Chapter I, 7.2]). Let $\eta : A \rightarrow A'$ be a homomorphism of rings with involution (Definition 1.0.6), and assume that A' is finitely generated and projective as a right A -module. Given a right A' -module M' , write $M'_{(A)}$ for the restriction of M' as a right A -module (via η). Let $\tau : A' \rightarrow A$ be a morphism of A -bimodules and assume that

$$\tau(\bar{a}) = \overline{\tau(a)}$$

for all $a \in A'$

The Scharlau transfer (Definition 1.6.3) map

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)})$$

(Definition 1.6.1) induces functors

$$T : \mathfrak{Herm}^{\varepsilon'}(A') \rightarrow \mathfrak{Herm}^{\varepsilon}(A)$$

$$T : \mathfrak{Herm}_+^{\varepsilon'}(A') \rightarrow \mathfrak{Herm}_+^{\varepsilon}(A)$$

(Definition 1.3.4)

It is appropriate to call these induced functors the *Scharlau transfer functor for Hermitian spaces over involution rings*.

Definition 1.6.5 ([?, Chapter I, (7.2.4) Proposition]). Let $\eta : A \rightarrow A'$ be a homomorphism of rings with involution (Definition 1.0.6).

An *involution trace* is a morphism $\tau : A' \rightarrow A$ of A -bimodules such that

- $\tau(ayb) = a\tau(y)b$ for all $a, b \in A$ and $y \in A'_{(A)}$, i.e. $\tau : A'_{(A)} \rightarrow A$ is a bimodule homomorphism.

-

$$\tau(\bar{a}) = \overline{\tau(a)}$$

for all $a \in A'$

- The 1-hermitian form $t_{A'/A} : A' \times A' \rightarrow A$ given by

$$t_{A'/A}(x, y) = \tau(\overline{xy})$$

is nonsingular (Definition 1.2.3).

Proposition 1.6.6 ([?, Chapter I, (7.2.4) Proposition]). Let $\eta : A \rightarrow A'$ be a homomorphism of rings with involution (Definition 1.0.6), and assume that A' is finitely generated and projective as a right A -module. Given a right A' -module M' , write $M'_{(A)}$ for the restriction of M' as a right A -module (via η).

Let $\tau : A' \rightarrow A$ be an involution trace (Definition 1.6.5).

The Scharlau transfer functors (Definition 1.6.4)

$$T : \mathfrak{Herm}^{\varepsilon'}(A) \rightarrow \mathfrak{Herm}^{\varepsilon}(A)$$

$$T : \mathfrak{Herm}_+^{\varepsilon'}(A) \rightarrow \mathfrak{Herm}_+^{\varepsilon}(A)$$

induced by τ in turn induce functors

$$T : \mathfrak{H}^{\varepsilon'}(A') \rightarrow \mathfrak{H}^{\varepsilon}(A)$$

$$T : \mathfrak{H}_+^{\varepsilon'}(A') \rightarrow \mathfrak{H}_+^{\varepsilon}(A)$$

(Definition 1.3.6) which may also be called *Scharlau transfer functors*.

1.7. Symmetric algebras with involution rings.

Definition 1.7.1 (Frobenius R -algebra). Let R be a ring and A be an R -algebra.

We say that A is a *Frobenius R -algebra* if either of the a posteriori equivalent conditions hold:

- A is a *left Frobenius R -algebra* if
 - A is finitely generated projective as a left R -module, and
 - there exists an isomorphism of left A -modules:

$$\psi_L : {}_A A \xrightarrow{\cong} {}_A A^* \quad \text{where } A^* = \text{Hom}_R(A, R)$$

with the action $(u \cdot f)(a) = f(au)$. This is equivalent to the existence of a left R -linear form $\varphi : A \times A \rightarrow R$ such that $b(x, y) = \varphi(xy)$ is non-degenerate (the map $x \mapsto \varphi(x \cdot)$ is an isomorphism). (♠ TODO: non-degenerate bilinear form)

- A is a *right Frobenius R -algebra* if
 - A is finitely generated projective as a right R -module, and
 - there exists an isomorphism of right A -modules:

$$\psi_R : A_A \xrightarrow{\cong} A_A^* \quad \text{where } A^* = \text{Hom}_R(A, R)$$

with the action $(f \cdot v)(a) = f(va)$. This is equivalent to the existence of a linear form φ such that $b(x, y) = \varphi(yx)$ is non-degenerate (the map $y \mapsto \varphi(\cdot y)$ is an isomorphism).

For an R -algebra A that is finitely generated and projective over R , these two definitions are equivalent. If a "left" isomorphism ψ_L exists, the corresponding trace $\varphi = \psi_L(1)$ can always be used to construct a "right" isomorphism ψ_R .

The difference between the two structures is encapsulated by the *Nakayama automorphism* $\alpha \in \text{Aut}(A)$. While a Symmetric Algebra (Definition 1.7.2) satisfies $\varphi(xy) = \varphi(yx)$ (implying $\psi_L = \psi_R$ as bimodules), a general Frobenius algebra only satisfies $\varphi(xy) = \varphi(y\alpha(x))$.

Definition 1.7.2 (cf. [?, Chapter I, 7.4] in the case that the base ring is commutative.). Let R be a not necessarily commutative ring.

A *symmetric R -algebra* is an R -algebra A such that:

1. A is finitely generated and projective as a one-sided R -module.
2. There exists an isomorphism of (A, A) -bimodules $\psi : A \xrightarrow{\cong} A^\vee$, where the dual (Definition 1.1.7) $A^\vee = \text{Hom}_R(A, R) = \text{Hom}_{R-\mathbb{Z}}(A, R)$ is the set of left R -linear maps.

The (A, A) -bimodule structure on A^* is defined for $f \in A^*$ and $u, v, a \in A$ by:

$$\begin{aligned}(u \cdot f)(a) &= f(au) \\ (f \cdot v)(a) &= f(va)\end{aligned}$$

Under this isomorphism, the element $\varphi = \psi(1)$ (the *symmetric trace*) necessarily satisfies $\varphi(ab) = \varphi(ba)$ for all $a, b \in A$.

A posteriori, the fact that A is a symmetric R -algebra implies that A is in fact projective as an R - R -bimodule.

Definition 1.7.3. Let R be an involution ring (Definition 1.0.5).

A symmetric R -algebra (Definition 1.7.2) A is said to be a *symmetric R -algebra with involution* if

- A is finitely generated and projective as an R -module,
- A has an R -linear involution (Definition 1.0.8), and
- the symmetric trace (Definition 1.7.2) $\varphi \in A^\vee$ (Definition 1.1.7) is an involution trace (Definition 1.6.5).

Definition 1.7.4. Let G be a group and R a not necessarily commutative ring.

The *group ring* $R[G]$, also denoted RG , is the set of all formal linear combinations of the form

$$\sum_{g \in G} r_g g$$

where $r_g \in R$ and $r_g = 0$ for all but finitely many $g \in G$. This set is endowed with a ring structure via:

- Addition:

$$\left(\sum_{g \in G} r_g g \right) + \left(\sum_{g \in G} s_g g \right) = \sum_{g \in G} (r_g + s_g) g$$

- Multiplication: Defined by the extension of the group law and the requirement that R and G commute ($rg = gr$):

$$\left(\sum_{g \in G} r_g g \right) \left(\sum_{h \in G} s_h h \right) = \sum_{k \in G} \left(\sum_{gh=k} r_g s_h \right) k$$

The identity element of $R[G]$ is $1_R 1_G$.

$R[G]$ may also be called the *group algebra of G over R* as the natural ring homomorphism $R \rightarrow R[G]$ sending $r \mapsto r 1_G$ is an algebra

Definition 1.7.5 (see [?, 7.3.7]). Let (R, σ_R) be a ring with involution (Definition 1.0.5). Let G be a group.

The *canonical involution on the group ring $R[G]$* (Definition 1.7.4) is the R -linear involution (Definition 1.0.8) given by

$$\overline{\sum r_g g} = \sum \sigma_R(r_g) g^{-1}.$$

If R is not specified with an involution structure, then usually, the canonical involution on $R[G]$ refers to the above in the case that R is equipped with the trivial involution (Definition 1.0.7).

Definition 1.7.6 (see [?, 7.3.7]). Let (R, σ_R) be a ring with involution (Definition 1.0.5). Let G be a group.

The *augmentation trace* or *standard trace* $\tau : R[G] \rightarrow R$ is defined as

$$\tau \left(\sum_{g \in G} a_g g \right) = a_1$$

where 1 is the identity element of G .

Assuming that R is commutative and that G is finite, this is an involution trace (Definition 1.6.5) with respect to the canonical involution (Definition 1.7.5) of $R[G]$ and is also a symmetric trace (Definition 1.7.2), and hence $R[G]$ is a symmetric algebra with involution (Definition 1.7.3) over R .

Example 1.7.7. The following are examples of symmetric algebras (Definition 1.7.2):

1. Finite field extensions (♠ TODO:)
2. Finite étale R -algebras (♠ TODO:)
3. Azumaya R -algebras (♠ TODO:)
4. Group rings (Definition 1.7.4) $R[G]$ equipped with the canonical involution (Definition 1.7.5) where R is commutative ring and a G is a finite group are symmetric algebras with involution (Definition 1.7.3).

Definition 1.7.8 (cf. [?, Chapter I. 7.4] in the case that the base ring is commutative). Let R be a not necessarily commutative ring, and A a symmetric R -algebra (Definition 1.7.2) with an R -linear involution (Definition 1.0.8) $\sigma : A \rightarrow A$ denoted by $\bar{a}$. Let N be a finitely generated projective right A -module. We view N also as an (R, A) -bimodule via the R -algebra structure of A .

An R -valued form $k : N \otimes_{\mathbb{Z}} N \rightarrow R$ (which is tantamount to a bilinear form $N \times N \rightarrow R$ (??)) is said to **admit A** if it is an (R, R) -bimodule map that factors through the tensor product over A . Specifically, it must satisfy:

1. **Balanced over A :** For all $x, y \in N$ and $a \in A$,

$$k(xa, y) = k(x, ay)$$

where the right action of A on the left side is the module action, and the left action of A on the right side is defined via the involution: $ay := y\bar{a}$.

2. **Compatibility with R -structure:** For all $r_1, r_2 \in R$,

$$k(r_1x, yr_2) = r_1k(x, y)r_2$$

This is equivalent to k inducing a well-defined (R, R) -bimodule homomorphism:

$$\tilde{k} : N \otimes_A \bar{N} \longrightarrow R$$

where $\bar{N}$ is the opposite (Definition 1.1.9) of N as a right module over the involution ring A .

Definition 1.7.9 (cf. [?, Chapter I, 7.4] for a discussion in the case where the base ring is commutative). Let R be a not necessarily commutative ring, let $\varepsilon \in R^\times$ satisfy $\varepsilon^2 = 1$, and A a symmetric R -algebra (Definition 1.7.2) with an R -linear involution (Definition 1.0.8).

The **category of ε -Hermitian spaces over R which admit A** is the full subcategory $\mathfrak{H}^\varepsilon(A)$ of the category of modules with ε -Hermitian forms over R (Definition 1.2.2) whose objects are ε -Hermitian spaces (Definition 1.3.5) over R whose underlying Hermitian forms admit A (Definition 1.7.8). In particular, such a category is a groupoid.

The **category of even ε -Hermitian spaces over R which admit A** is the full subcategory $\mathfrak{H}_+^\varepsilon(A)$ whose objects are those of $\mathfrak{H}^\varepsilon(A)$ whose underlying Hermitian forms are even (Definition 1.2.6).

Theorem 1.7.10 ([?, Chapter I, 7.4.1 Theorem]). Let R be a commutative ring, and let A be a symmetric R -algebra with involution (Definition 1.7.3).

The Scharlau transfer functors (Definition 1.6.4)

$$T : \mathfrak{Herm}^\varepsilon(A) \rightarrow \mathfrak{Herm}^\varepsilon(R)$$

$$T : \mathfrak{Herm}_+^\varepsilon(A) \rightarrow \mathfrak{Herm}_+^\varepsilon(R)$$

give rise to isomorphisms of categories

$$\mathfrak{H}^\varepsilon(A) \xrightarrow{\sim} \mathfrak{H}^\varepsilon(R; A)$$

$$\mathfrak{H}_+^\varepsilon(A) \xrightarrow{\sim} \mathfrak{H}_+^\varepsilon(R; A)$$

(Definition 1.3.6) (Definition 1.7.9) which preserve orthogonal sums and hyperbolic spaces.

(♠ TODO: orthogonal sums, hyperbolic spaces)

1.8. Involution, sesquilinearity, hermitian form over sheaves of rings.

Definition 1.8.1. Let $(\mathcal{S}, \tau)$ be a site. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{S}$. An *involution on $\mathcal{O}$* is a morphism of sheaves of rings $\sigma : \mathcal{O} \rightarrow \mathcal{O}^{\text{op}}$ such that $\sigma \circ \sigma = \text{id}_{\mathcal{O}}$, i.e. an involution (Definition 1.0.3) in the category of sheaves of rings on $\mathcal{S}$.

Definition 1.8.2. Let $(\mathcal{S}, \tau)$ be a site. Let $(\mathcal{O}, \sigma)$ be a sheaf of rings on $\mathcal{S}$ with involution (Definition 1.8.1) $\sigma : \mathcal{O} \rightarrow \mathcal{O}$. Let $\mathcal{E}$ be a sheaf of left $\mathcal{O}$ -modules on $\mathcal{S}$.

A *sesquilinear form on $\mathcal{E}$* is a morphism of sheaves of abelian groups

$$\phi : \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{O}$$

such that for every object $U \in \mathcal{S}$, the induced map on sections

$$\phi_U : \mathcal{E}(U) \times \mathcal{E}(U) \rightarrow \mathcal{O}(U)$$

is a σ_U -sesquilinear form (Definition 1.1.1) on the $\mathcal{O}(U)$ -module $\mathcal{E}(U)$, where $\sigma_U : \mathcal{O}(U) \rightarrow \mathcal{O}(U)$ is the involution (Definition 1.0.5) on global sections.

Definition 1.8.3. Let $(\mathcal{O}, \sigma)$ be a sheaf of rings with involution (Definition 1.8.1) on a site $\mathcal{S}$. Let $\mathcal{E}$ be a sheaf of left $\mathcal{O}$ -modules.

A *hermitian form on $\mathcal{E}$* is a morphism of sheaves of sets

$$\phi : \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{O}$$

such that for every object U in $\mathcal{S}$, the map on sections

$$\phi_U : \mathcal{E}(U) \times \mathcal{E}(U) \rightarrow \mathcal{O}(U)$$

is a hermitian form (Definition 1.2.1) on the $\mathcal{O}(U)$ -module $\mathcal{E}(U)$ with respect to the involution (Definition 1.0.5) $\sigma_U : \mathcal{O}(U) \rightarrow \mathcal{O}(U)$.